

Muhammad Imran, PhD

Clinical Medical Physics Resident – Diagnostic Imaging

Clinical Profile

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Diagnostic medical physics resident with hands-on experience in quality control and performance evaluation of radiography, fluoroscopy, CT, MRI, mammography, ultrasound, and nuclear medicine systems. Background in quantitative image-quality analysis, radiation safety, clinical research, artificial intelligence, and scientific computing.

Education

2025–2027	Clinical Medical Physics Residency , University of Chicago
2024–2025	Certificate in Medical Physics , Florida Atlantic University
2020–2024	Postdoctoral Research , University of Nevada, Reno & Iowa State University
2013–2020	PhD in Physics , University of Florida

Clinical Medical Physics Experience

Diagnostic Medical Physics Resident

2025–2027

University of Chicago

- Performed annual performance evaluations, quality control, and acceptance testing of diagnostic imaging systems, including radiography, fluoroscopy, CT, MRI, ultrasound, mammography, and nuclear medicine.
- Performed radiation shielding design calculations for a diagnostic radiography room using NCRP methodology, including workload, use factor, occupancy, design goal, and barrier transmission requirements.
- Gained experience with CT protocol development and optimization principles, including acquisition parameters, image quality, radiation dose, and clinical considerations.
- Performed peak skin dose calculations for fluoroscopic and interventional procedures using radiation dose and procedural data.
- Observed clinical Y-90 radioembolization and Pluvicto (Lu-177) cases as part of nuclear medicine and radiation safety training.
- Reviewed imaging practices and quality assurance activities for compliance with University of Chicago Medicine policies, ACR requirements, AAPM recommendations, and applicable regulatory requirements.

Mobile C-Arm Image Quality Research

- Evaluated image quality across 11 mobile C-arm fluoroscopy systems using a contrast-detail phantom under standardized imaging conditions.
- Compared human observer detectability with quantitative Rose signal-to-noise ratio (Rose-SNR) analysis and generated contrast-detail curves for system performance ranking.

AI-Based Mammography Quality Control

- Learned and implemented an FDA-developed deep learning software package for automated image quality assessment of digital breast tomosynthesis (DBT) using the CDMAM 4.0 phantom.
- Gained hands-on experience with the analysis workflow, including ROI extraction, software execution, interpretation of performance metrics, and quality assurance reporting.
- Utilized Python- and ImageJ/FIJI-based tools for image preprocessing and automated ROI extraction as part of the AI quality control pipeline.

Diagnostic Imaging Equipment Experience

Modality / Equipment	Vendor	Model
Radiography	Philips	DigitalDiagnost
	GE	Discovery XR656 G2
	Shimadzu	MobileDaRt MUX-200D-XC, RADspeed Pro
Fluoroscopy / R&F	Philips	EasyDiagnost
	Siemens	Luminos Agile Max
Interventional Radiology / Angiography	Philips	Azurion 7 B20/15
	Canon	Alphenix INFX-8000C
Mobile C-Arm CT	GE	OEC Elite
	GE	Revolution Apex, Revolution CT, Revolution GSI
	Philips	Brilliance iCT 256-slice
Mobile CT	NeuroLogica	OmniTom NL 5000, BodyTom NL 4100
Dental / Cone-Beam CT	Carestream Health	CS 9600
MRI	Philips	Achieva 3T, Ambition S 1.5T, Ingenia Elition X 3T
	Hologic	Selenia 3Dimensions
	Hologic	Affirm SBB add-on
Mammography Review Workstation	Barco	Coronis Uniti MDMC-12133
Specimen Radiography	Faxitron	Faxitron Core
DEXA / Bone Densitometry Ultrasound	Hologic	Horizon C
	Philips	EPIQ 7G, EPIQ Elite
	GE	LOGIQ E9
Automated Breast Ultrasound (ABUS)	Zonare	ZS3
	GE	Invenia ABUS
	Siemens	Biograph mCT, Biograph Vision 600
PET/CT	Philips	BrightView X
SPECT	GE	Discovery NM/CT 670 Pro, Discovery NM/CT 870 CZT
SPECT/CT	Capintec	CRC-55tR
Dose Calibrator	Capintec	CRC-55W
Well Counter / Well System	Capintec	CRC-55t
Hot-Lab / Well & Probe System		

Technical Skills

Clinical & Imaging	Radimetrics, RadMachine, DICOM, ImageJ/FIJI
Programming	Python, MATLAB, Mathematica, C++, Fortran
Data & AI	TensorFlow, Scikit-Learn, Pandas, NumPy
Scientific Computing	COMSOL, parallel programming, MPI-Python
Development	Git, Linux, Windows, macOS

Selected Publications & Scientific Presentations

M. Imran, I. Reiser, and Z. F. Lu, “Ranking of Mobile C-Arm Image Quality Using Contrast–Detail Detectability Metrics,” *AAPM-COMP Joint Annual Meeting*, Vancouver, Canada (2026).
Author/co-author of 9 peer-reviewed publications in physics journals.

Professional Meetings

2026	AAPM-COMP Joint Annual Meeting, Vancouver, Canada
2025	RSNA Annual Meeting, Chicago, Illinois